Homework #5

This is the homework for week 5. The due date is 11:59 p.m., October 20. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (20 points)

For a public opinion poll for a close presidential election, let p denote the proportion of voters who favor candidate A. How large a sample should be taken if we want the maximum error of the estimate of p to be equal to

(a) 0.03 with approximate 95% confidence?

(b) 0.02 with approximate 95% confidence?

(c) 0.03 with approximate 90% confidence?

(d) Since we do not know p in practice, please provide a valid upper bound for the number of samples in aforementioned problems which is independent of p. (Hint: you may use the fact that

$$\sqrt{p(1-p)} \leq \frac{p+1-p}{2} = \frac{1}{2}.$$

)

Answer: Suppose the observations are $X_1, \cdots, X_n$ and the estimator used for building confidence interval is

$$\frac{\sqrt{n}|\widehat{p}-p|}{\sqrt{\widehat{p}(1-\widehat{p})}} \leq z_{\alpha/2} \Rightarrow |\widehat{p}-p| \leq \frac{\sqrt{\widehat{p}(1-\widehat{p})}}{\sqrt{n}} \times z_{\alpha/2}.$$

(a) We have

$$\frac{\sqrt{\widehat{p}(1-\widehat{p})}}{\sqrt{n}} \times z_{\alpha/2} \leq 0.03 \Rightarrow n \geq \frac{\widehat{p}(1-\widehat{p})}{0.03^2} \times 1.98^2.$$

(b) Similarly,

$$\frac{\sqrt{\widehat{p}(1-\widehat{p})}}{\sqrt{n}} \times z_{\alpha/2} \leq 0.02 \Rightarrow n \geq \frac{\widehat{p}(1-\widehat{p})}{0.02^2} \times 1.98^2.$$

(c) Similarly,

$$\frac{\sqrt{\widehat{p}(1-\widehat{p})}}{\sqrt{n}} \times z_{\alpha/2} \leq 0.03 \Rightarrow n \geq \frac{\widehat{p}(1-\widehat{p})}{0.03^2} \times 1.65^2.$$

(d) We use $\sqrt{\widehat{p}(1-\widehat{p})} \leq \frac{1}{2}$. In that case, the first situation becomes 1089, the second situation becomes 2451, the third situation becomes 756.

(**Question2):** (30 points)

Suppose $X_1, X_2, \cdots, X_n$ are i.i.d. Poisson random variables with parameter λ, i.e.,

$$Prob(X_1 = k) = \frac{\lambda^k}{k!} \exp(-\lambda) \text{ for } k \geq 0.$$

Recall the MLE for λ is $\frac{1}{n}\sum_{i=1}^n X_i$.

(a) Derive the Fisher information.

(b) Provide a valid estimator for the Fisher information. You do not need to prove the consistency of your estimator.

(c) Derive an $1-\alpha$ confidence interval for λ.

(d) Suppose we want the maximum error of the estimate of λ to equal 0.01 with approximate 95% confidence, determine the sample size. Your result can include λ.

Answer: (a) The Fisher information function is

$$I_n(\lambda) = -\mathbf{E}\frac{\partial^2 \ell}{\partial \lambda^2} = \lambda^{-1} \times n.$$

(b) Since MLE for λ is $\widehat{\lambda} = \frac{1}{n}\sum_{i=1}^n X_i$, a good estimator for the Fisher information is

$$\widehat{I_n(\lambda)} = \frac{n}{\widehat{\lambda}}.$$

(c) From the central limit theorem, we have

$$\frac{\sqrt{n}}{\sqrt{\widehat{\lambda}}}(\widehat{\lambda} - \lambda) \to_d N(0,1),$$

so the $1-\alpha$ CI for λ is

$$\left[\widehat{\lambda} - \frac{\sqrt{\widehat{\lambda}}}{\sqrt{n}} z_{\alpha/2}, \widehat{\lambda} + \frac{\sqrt{\widehat{\lambda}}}{\sqrt{n}} z_{\alpha/2}\right].$$

(d) The sample size is

$$\left(\frac{\sqrt{\widehat{\lambda}}}{0.01} \times z_{0.025}\right)^2$$

(**Question3):** (30 points)

Suppose $X_1, \cdots, X_n$ are i.i.d. normal random variables following $N(0, a)$ distribution (mean 0, variance a).

(a) Derive the maximum likelihood estimator for a.

(b) Derive the Fisher information.

(c) Provide a valid estimator for the Fisher information. You do not need to prove the consistency of your estimator.

(d) Derive an $1-\alpha$ confidence interval for a.

Answer: (a) For the log–likelihood function is

$$\ell(a) = -\frac{\sum_{i=1}^n X_i^2}{2a} - \frac{n}{2}\log(2\pi a),$$

the MLE is $\widehat{a} = \frac{1}{n}\sum_{i=1}^n X_i^2$.

(b) The Fisher information is

$$I_n(a) = -\mathbf{E}\frac{\ell^2(a)}{\partial a^2} = -\frac{n}{2a^2} + \frac{1}{a^2}\sum_{i=1}^n \mathbf{E}X_i^2 = \frac{n}{2a^2}.$$

(c) A valid estimator for the Fisher information is

$$\widehat{I(a)} = \frac{n}{2\widehat{a}^2}.$$

(d) The $1-\alpha$ CI for a is

$$\left[\widehat{a} - \sqrt{\frac{2\widehat{a}^2}{n}} \times z_{\alpha/2}, \widehat{a} + \sqrt{\frac{2\widehat{a}^2}{n}} \times z_{\alpha/2}\right].$$

2 Programming Questions

(20 points)

Implement the following experiment using Jupyter or R:

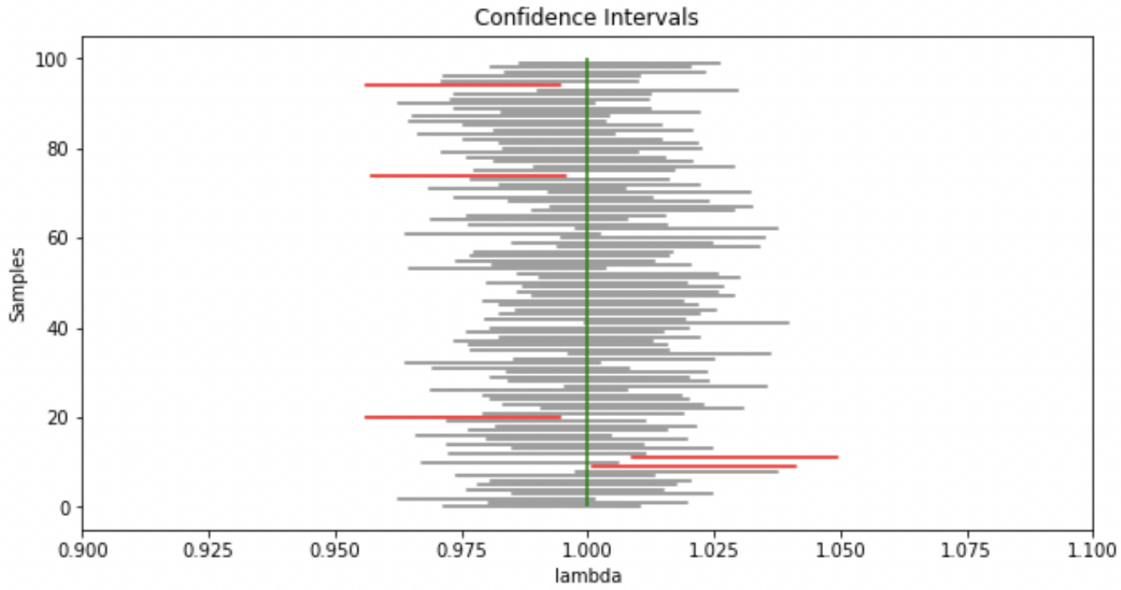
i. Generate $X_1, \cdots, X_n$ satisfying Bernoulli distribution with parameter $p = 0.3$ and $n = 40$.

ii. Plot the 95% confidence interval for p, i.e.,

$$\left[\widehat{p} - \frac{\sqrt{\widehat{p}(1-\widehat{p})}}{\sqrt{n}} \times z_{2.5\%},\ \widehat{p} - \frac{\sqrt{\widehat{p}(1-\widehat{p})}}{\sqrt{n}} \times z_{97.5\%}\right],$$

where z_α is the α upper quantile of a standard normal random variable.

iii. Repeat step i and ii for 100 times, see how many confidence intervals cover the real p. Your result should look like this



Please ignore the x–axis and y–axis scales of the figure, we just show what the result should look like.